

Sakurako Kingsbury

Hoboken, NJ | +1(914) 316-9828 | contact.kingsbury@gmail.com

Portfolio: sakurakokingsbury.com | LinkedIn: linkedin.com/in/sakurakokingsbury/

SKILLS

CAD & Simulation: SOLIDWORKS, Fusion 360, ANSYS

Programming: MATLAB, C++, Python, OpenCV

Technical: KiCad, Embedded Systems, Circuit & PCB Design, Microcontrollers, Sensor Integration, Motor Control, Soldering

Languages: English (Native), Japanese (Native)

EDUCATION

Stevens Institute of Technology | B.E. Mechanical Engineering

Hoboken, NJ | **Sept 2023 - Expected May 2027**

- Current GPA: 3.53/4.00
- Honors: Dean's List, Edwin A. Stevens Scholarship, Martha Bayard Stevens Scholarship

EXPERIENCE

Stevens Institute of Technology | Student Technical Staff

Hoboken, NJ | **Sept 2026 - Present**

Mechanical Engineering Department

- Support daily operations of Mechanical Engineering computer and design laboratories, including workstation maintenance and engineering software support.
- Assist students and faculty with technical issues involving engineering workstations, CAD software, and laboratory resources.

RAPID Institute | Mechatronics Research Intern

Dublin, Ireland / Continued Remotely | **June 2026 - Present**

Research Advances in Personalized Integrated Diagnostics (RAPID) Institute at Dublin City University

- Developed mechatronic systems and experimental equipment supporting centrifugal microfluidics and biomedical research.
- Developed a high-speed laser positioning system by integrating an ESP32-based controller, custom PCB, and galvanometer hardware for programmable beam control.
- Designed and prototyped a portable colorimetric sensing device integrating embedded cameras, controlled illumination, and image processing for biomedical fluid monitoring.

TS Imagine | Product & Systems Engineering Intern

Manhattan, NY | **June 2025 - June 2026**

- Configured and deployed 40+ enterprise workstations, including hardware selection and installation, BIOS/TPM setup, OS imaging, and system provisioning.
- Repaired and maintained data center hardware including PDU systems to ensure operational reliability.
- Built and maintained Jira Assets infrastructure tracking hundreds of devices across engineering and business teams.

SELECTED PROJECTS

[Chu Chu: An Interactive Companion Robot](#)

- Served as electromechanical lead, designing the complete robot assembly, including the drivetrain, structural components, and actuator layout to support autonomous navigation, responsive interactions, and expressive motion.
- Architected the electrical system layout by defining subsystem placement in SOLIDWORKS, developing the wiring schematic, and integrating the five sensing, actuation, and feedback subsystems into a microcontroller-based control system, enabling over seven hours of autonomous and user-driven operation.
- Validated mechanical design decisions through structural, torque, and power analyses prior to fabrication, eliminating redesign during assembly.

[NaviBot: Autonomous Navigator](#)

- Completed the 36" x 96" autonomous obstacle course in 1 min 21 sec with zero collisions by combining ultrasonic sensing, real-time path planning, and motor control.
- Optimized chassis size and weight by designing a compact integrated structure with thin-walled features that housed the drivetrain, sensors, and electronics while maintaining structural integrity and serviceability.
- Implemented real-time obstacle avoidance logic in C++ by fusing data from three 45° ultrasonic sensors with LiDAR-based MQTT localization, enabling autonomous collision-free navigation.
- Rapidly iterated six 3D-printed chassis designs over a 5-week development cycle, refining sensor integration, electronics packaging, structural reliability, and manufacturability through repeated testing to achieve reliable autonomous performance.